

Bio-Inspired Morphing Wing Design

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I. PROBLEM STATEMENT AND MOTIVATION

Conventional aircraft and UAV wings rely on discrete control surfaces such as flaps and ailerons to manipulate airflow and generate lift or torque. Although effective, these mechanisms introduce geometric discontinuities that cause flow separation and increased drag, which is especially problematic at low Reynolds numbers of small UAVs. In addition, actuator-based systems add mass, complexity, and failure points.

In contrast, birds modulate aerodynamic surfaces through continuous wing shape changes. They adjust camber, span, and twist in real time without discrete control surfaces. Swifts and falcons, for example, actively control wing curvature and twist to optimize lift-to-drag ratio, delay stall, and respond to gusts [1]. These biologically inspired strategies suggest a pathway for low-mass, high-agility UAVs.

This project explores such avian-inspired morphing mechanisms, beginning with simulating camber deformation in a 2D elastic beam. The beam represents a chordwise wing cross-section that passively or actively changes shape via compliant materials. This simplified model allows for studying how boundary actuation, like tip displacement or angle, can induce internal curvature.

Ultimately, this framework can extend to twist and 3D shell models that capture full-span morphing. By simulating feedback-controlled compliant systems, we aim to understand how simple mechanisms might replicate the agility of bird flight without complex actuation or high-fidelity aeroelastic modeling.

II. BACKGROUND AND LITERATURE REVIEW

A. Biological Inspiration

Birds and bats achieve remarkable agility through continuous geometric transformations. They dynamically adjust wing span, camber, and twist to optimize flight performance across maneuvers, energy efficiency, and gust rejection. Lentink et al. [1] showed that swifts actively control camber during gliding to enhance lift and stability. These deformations are achieved through musculoskeletal structures and passive material compliance, rather than discrete joints or actuators.

B. Engineering Implementations

Traditional morphing wings use embedded actuators, hinges, and linkages [2], which increase weight and introduce failure points. To mitigate this, researchers have explored smart materials, such as shape memory alloys and piezoelectrics, that

allow distributed deformation with fewer moving parts [3]. These materials are particularly attractive for low-Reynolds-number UAVs where drag from traditional control surfaces is prohibitive.

Beam and plate models have been used as simplified morphing representations [4]. Bending-based deformation simulates camber, while torsion models represent twist. However, most work assumes either passive response to loads or complex embedded actuation.

C. Control Challenges

Controlling compliant structures is nontrivial due to their nonlinear, underactuated nature. Prior studies have used PID and model predictive control to regulate shape in soft robots and morphing airfoils [5], but these approaches often require full-state sensing or complex modeling, which is impractical for lightweight UAVs.

III. PROPOSED APPROACH AND CONTRIBUTIONS

A. Modeling Strategy

We model a chordwise cross-section of a morphing wing as a 2D elastic beam using lumped-mass discretization and Euler–Bernoulli beam theory. The model captures axial stiffness (EA), bending stiffness (EI), inertia, and gravitational loading. A fully implicit time integrator combined with a Newton–Raphson solver is used for stable evolution over time under large deformations.

Boundary conditions are applied as follows:

- Node 0 is fully fixed, anchoring the root of the beam.
- Node 1 has controlled positional constraints that can be actuated to modulate beam curvature and thereby induce camber.
- Node N (tip) is left free to deform passively in response to boundary actuation.

This asymmetric constraint strategy mimics the structural layout of bird feathers, where feathers are anchored near the wing root but free along their trailing edges, allowing shape modulation with minimal actuation.

B. Control Framework

To replicate bio-inspired shape change, we implement a proportional-integral (PI) feedback controller that tracks a desired trajectory of the beam’s midpoint and tip orientation. The control inputs correspond to small displacements of node 1 in both x and y , as well as adjustments to the angle between

nodes 1 and 2. We will define a time-varying curvature target $\theta^*(t)$ to simulate dynamic lift modulation or changing wing camber. Control commands are saturated to reflect physical limits of small UAV actuators, preserving feasibility and preventing numerical instability.

C. Material and Structural Properties

Beam parameters are chosen to approximate the flexible skin or spar of a lightweight unmanned aerial vehicle (UAV) wing. Material stiffnesses, density, and geometric scaling are selected such that the structure is compliant enough to achieve noticeable curvature under modest boundary displacement.

We also plan to investigate non-uniform and potentially non-linear stiffness distributions to explore how passive material gradients might enable tunable shape profiles. Future versions may incorporate drag forces dependent on local camber and orientation.

D. Anticipated Contributions

This project aims to contribute:

- A reduced-order simulation framework for 2D compliant morphing with biologically plausible boundary control.
- A method for tracking midspan position and camber angle using minimal actuation and feedback control.
- A flexible modeling setup that can be extended to plate and shell elements for future 3D camber-twist morphing studies.
- Insights into passive and active material distributions that achieve desired wing shapes with minimal sensing and actuation.

This framework forms a stepping stone between simple beam models and high-fidelity aeroelastic simulations, emphasizing structural actuation over aerodynamic modeling.

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